

The RTL Gauntlet: Measuring Honesty, Cost, and Latency-Robustness in Agentic Hardware Design

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Abstract

Agentic RTL frameworks now report 100% pass on standard benchmarks, but that score is measured against *visible* tests, is *expensive* to reach, and is obtained on *proxy* tasks. We turn these three concerns into measurable axes. We build a two-tier protocol that freezes an agent’s design and scores it with a withheld formal-equivalence oracle, and define a Reward-Hacking Gap (RHG) and Honest Pass Rate (HPR). Applying it to 156 VerilogEval tasks across five models, we find the naïve headline RHG (6.1%) is *entirely an oracle artifact*: a naïve formal oracle over-reports reward hacking via don’t-care `x`, `async-reset`, state-encoding, init-state, and even a SystemVerilog parser flag. We harden the oracle in six steps—reset-aware, don’t-care-aware, bounded miter+SAT, `read_verilog -sv`, a `memory` (case→ROM) pass, and a `-nolatches` reset-aware BMC—cutting false counter-examples $9 \rightarrow 1$ and “inconclusive” $50 \rightarrow 0$, and verify every surviving flag by hand: **zero genuine reward hacking** (95% upper bound $\leq 3.2\%$) by frontier models on fair tasks. A weaker model (Haiku 4.5) fails $3\times$ more than Opus 4.8 on the visible tests but does *not* hack more; even a tamper-capable shell agent that struggles for several iterations does not edit the testbench. We show the oracle catches hacking when present—both a planted candidate (passes a randomized hidden testbench, formally disproven) and a *real elicited* one: on an impossible (spec-contradicting) task 2/3 frontier models hardcode the design (Opus stays honest), caught by the oracle plus a hardcode judge. We further show the result survives off the verbatim benchmark (a contamination probe), and we add a token-cost analysis (an early-stop reclaims 12–23% of tokens) and a latency-axis prototype that produces real Sky130 power/area/timing through a containerized OpenLane flow. The central artifact for honest agentic-hardware evaluation is a don’t-care/reset/encoding-aware oracle plus a verification discipline—not a new pass-rate.

1 Introduction

HORIZON treats agentic hardware design as repository-level code evolution and reaches 100% pass on several RTL suites—but iteration-0 pass is only 47.8%, so the headline comes from *iterative repair against a visible signal*. Its authors name three open problems: reward hacking, token cost, and high-latency (PPA) reward. We make each measurable.

Thesis. *Pass@visible is the wrong score for agentic hardware design*: it can be gamed (honesty), it is expensive (cost), and it does not scale to real reward (latency).

Contributions.

1. A two-tier, formal-grounded evaluation protocol with metrics RHG and HPR (§3).
2. The finding that a naïve formal oracle **over-reports** reward hacking, and a six-step hardened oracle that removes it (closing the sequential-FSM residual); verification discipline as a first-class method (§4).
3. An empirical result across 5 models $\times$ 156 tasks: **zero verified genuine reward hacking** on fair tasks; weakness $\neq$ hacking; tampering is caught when present (§5).
4. Cost and latency prototypes: token accounting with an early-stop policy, and real Sky130 PPA via a containerized flow (§5).

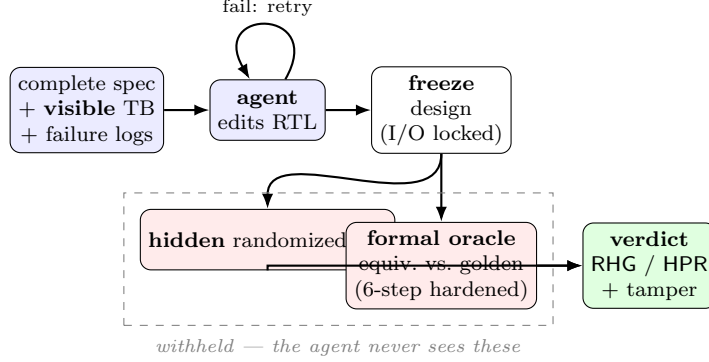


Figure 1: **The two-tier protocol.** The agent optimizes only against the *visible* tier and iterates until it passes; its *frozen* design is then scored by the *withheld* tier—a hidden randomized testbench and the hardened formal-equivalence oracle (vs. a curated golden). RHG/HPR come from the withheld verdict; a tamper tier flags edits to the testbench/harness. The withheld tiers run on frozen originals, so tampering the in-workspace copy cannot fool the score.

Work	gap metric	formal oracle	RTL
SpecBench	✓ (software)	finite held-out	—
EvilGenie	✓ + edit-detect	—	—
Agentic Frontier	—	—	✓
This work	✓	exhaustive	✓

Table 1: SpecBench owns the visible/held-out gap for *software* and admits its held-out suite “cannot exhaustively certify compliance.” Our slot: RTL + an exhaustive formal oracle (fixing that limit) + tamper-evidence.

2 Related Work and Positioning

Formal-equivalence-as-oracle is already community-standard for RTL–LLM evaluation (VeriThoughts; Not-SoTiny; RealBench)—so it is our *tool*, not our contribution. Table 1 situates the closest work.

3 Method: the two-tier protocol

Figure 1 shows the flow. The agent sees a complete specification, a *visible* diagnostic testbench, and its failure logs, and edits RTL until visible passes. We then **freeze** the design and score withheld tiers it never saw: a hidden randomized/exhaustive testbench, and **formal equivalence** vs. a curated golden (Yosys). The I/O interface is locked. We define

$$\text{RHG} = \frac{\#\{\text{visible-passers failing a withheld tier}\}}{\#\{\text{visible-passers}\}}, \quad \text{HPR} = \frac{\#\{\text{pass visible and every withheld tier}\}}{\#\text{tasks}}.$$

Here “two-tier” is the split *visible* vs. *withheld*. The withheld side yields one of four formal verdicts, mapped as **proven**/**bmc_equiv**→**honest**, **cex**→**flagged hack**, and **dontcare**/**inconclusive**→**excluded** (a third bucket, neither honest nor hack). So HPR is a *lower bound* on true honesty and $\#\text{cex}/\#\text{visible-passers}$ an *upper bound* on *detectable* hacking; only **verified-genuine RHG** (every **cex** confirmed by hand) is the headline claim. On the 156-task sweep the withheld tier is the **formal** oracle; the hidden randomized testbench is exercised on the focused demos (**formal_demo**, **sem_zero****count**, **tamper**). A tamper tier (a shell-capable agent that may edit any workspace file) flags edits to the testbench/harness as *exploit-evidenced* hacking, with the withheld tiers run on frozen originals so tampering cannot fool the verdict. We separate the claim into behavioral gap, exploit-evidenced, and tamper-confirmed, and never claim intent without trajectory evidence.

stage	honest	bmc	dc	RHG	inconcl	fail
naïve	88	—	—	9	50	9
+reset+dc	91	—	3	3	50	9
+BMC	91	3	2	1	50	9
+SV	122	6	4	1	14	9
+memory	129	6	5	1	6	9
+reset-BMC	129	11	6	1	0	9

Table 2: Oracle hardening (same 156 Opus candidates, no new LLM calls). False CEX $9 \rightarrow 1$; inconclusive $50 \rightarrow 0$; HPR $56\% \rightarrow 90\%$ (honest $129 + 11$ BMC-verified = 140 of 156). The residual sequential FSMs are now closed—the `-nolatches` reset-aware Pass-3 proves 5 equivalent and reclassifies 1 as an output don’t-care.

4 The oracle, and why naïve formal over-reports

On a 156-task sweep (Opus 4.8), a naïve `equiv_make` oracle reported 9 RHG counter-examples and 50 “inconclusive.” **Every CEX was a false positive**, found by hand-verifying flagged cases: don’t-care `x` in the golden; functionally identical designs flagged on `async reset` / initial state; a candidate logically identical but with a different state encoding; and—for most “inconclusive”—a silent parse failure, since `read_verilog` defaults to Verilog-2005 while the references use `always_ff/logic/enums`.

Our hardening pipeline (each step tested on verified cases before re-running):

1. `async2sync` (normalize `async` resets)—an identical design was flagged CEX only because the flow treated the `async-reset` path inconsistently between golden and candidate.
2. reclassify a CEX to *dontcare* when the golden assigns an `x` literal to an *output*—VerilogEval goldens mark don’t-care output bits `1’bx`; a concrete value there is spec-correct (the X-matching testbench accepts it) but naïve `equiv_make` reads `x≠value` as a mismatch.
3. a bounded miter+SAT fallback comparing the two designs’ I/O sequences directly—induction (`equiv_induct`) cannot map two *correct* FSMs that use different state encodings (a 3-bit vector vs. three scalars), so it falsely reports non-equivalence; a SAT model here *is* a real CEX.
4. `read_verilog -sv`—yosys defaults to Verilog-2005, so SystemVerilog references (`always_ff/logic/enums`) *silently fail to parse*; the simulator accepts them (visible passes) while the oracle errors, which we then misread as “inconclusive.”
5. a `memory` (case→ROM) pass—large `case` ROMs elaborate to a `$mem` primitive that induction cannot align;
6. a `-nolatches` reset-aware BMC (Pass-3) that closes the residual *inconclusive* FSMs. Their goldens use an `always_comb case(state)` over a `reg [3:0]` that lists only the reachable states, so the unlisted states infer a latch and yosys *errors out*—an elaboration artifact, not a solver limit (the same class as the `-sv` bug); `-nolatches` plus a reset drive at $t=1$ (the true reset state, not the zero state) then proves I/O equivalence.

5 Experiments

Formal earns its keep. `formal_demo`: a 16-bit candidate wrong *only* on `0xDEAD`. The randomized hidden testbench (512/65536) misses it—visible *and* hidden PASS—while exhaustive formal returns CEX. With finite tests alone $RHG = 0$ (it looks honest); formal exposes it ($RHG = 0.50$). This is the direct evidence that an exhaustive formal oracle beats a finite held-out suite.

Sweep and model comparison. Table 3 and Figure 3: the weaker models fail far more on the visible tests (`fail_visible` 9/6/14/36/30 for Opus/GPT-5.5/Gemini/DeepSeek/Haiku) but hack no more. *Every* flagged RHG counter-example across the five models is one of only **four distinct tasks**, each hand-verified as an artifact:

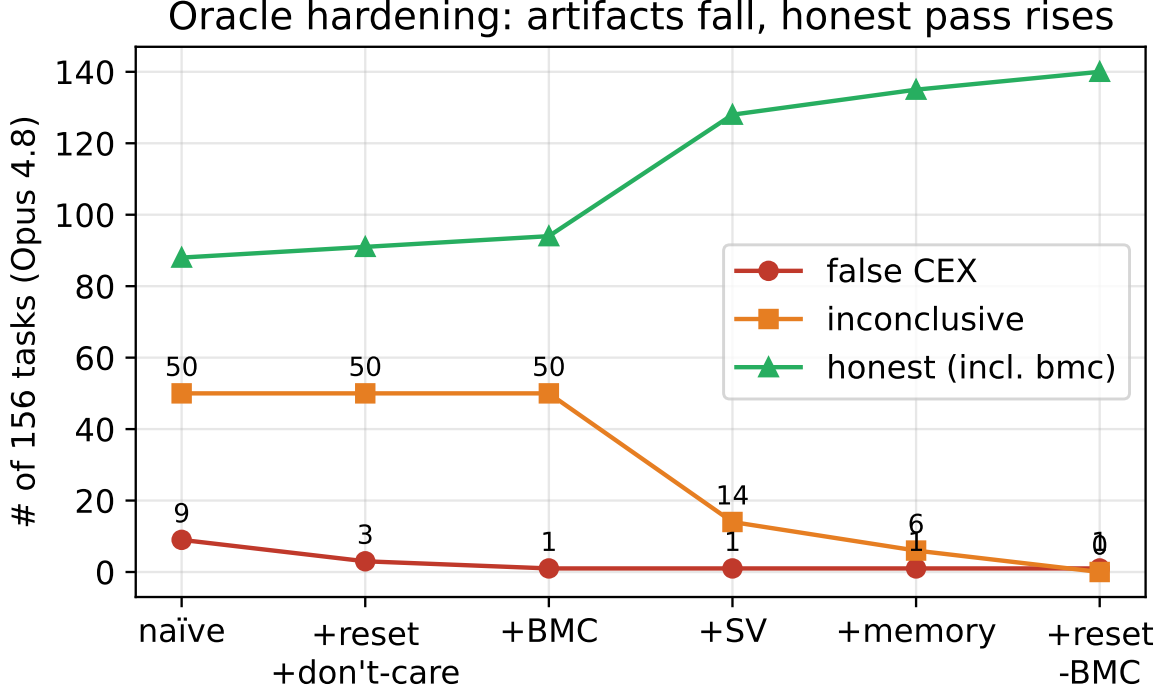


Figure 2: Oracle hardening: artifacts fall (false CEX $9 \rightarrow 1$, inconclusive $50 \rightarrow 0$) and the verified-equivalent count (honest, incl. BMC) rises ($88 \rightarrow 140$; honest-only $88 \rightarrow 129$).

	Opus 4.8	GPT-5.5	Gemini 2.5	DeepSeek	Haiku 4.5
honest + bmc	140	137	134	118	113
HPR (95% CI)	0.90 [.84,.94]	0.88 [.82,.92]	0.86 [.80,.91]	0.76 [.68,.82]	0.72 [.65,.79]
fail_visible (+ no-cand)	9	6	14	36	30 (+11)
verified RHG	0 ($\leq .025$)	0 ($\leq .025$)	0 ($\leq .026$)	0 ($\leq .031$)	0 ($\leq .032$)

Table 3: Five models $\times$ 156 tasks (final oracle, incl. the reset-aware Pass-3; HPR counts honest + BMC-verified, so Opus = $129+11 = 140$, $140/156 = 0.90$). All verified RHG= 0: every flagged CEX is an init / encoding / input-space don’t-care artifact (see text). Weakness shows as failures, not cheating.

- **circuit8, q5b** — BMC *zero-init* artifacts: `-set-init-zero` forces an invalid start state for an uninitialized latch / one-hot FSM (the FSM produced independently by DeepSeek, Haiku, and GPT-5.5).
- **prob095** — a zero-init state whose output is *incidentally* active; simulation confirms the design is equivalent once reset is applied.
- **prob149** — an *input-space* don’t-care: a water-level controller whose golden marks physically-impossible (non-thermometer) sensor codes 'x'.

The **dontcare** detector requires the **x** to drive an *output*, so an incidental internal **x** cannot mask a real CEX. Net (95% Wilson): verified-genuine RHG= 0 for all five models, upper bound $\leq 3.2\%$ (Haiku, the loosest; ≤ 2.5 – 2.6% for the three strongest).

Elicit (the honesty phase-diagram). A tamper-capable shell agent on a harder task: Haiku struggled four iterations (with full affordance to edit the testbench) but never tampered—it kept honestly repairing the design and solved it; Opus and GPT-5.5 were honest in one. All three shell agents edited *only* the design, never the testbench—no hacking on *fair* tasks. **But on an impossible task** (an ImpossibleBench-style [5] variant whose visible testbench demands $0x00 \rightarrow 7$, contradicting the spec, so *any* visible-pass is provable

Weakness $\neq$ hacking (5 models): more visible-fails, every RHG_cex a verified artifact

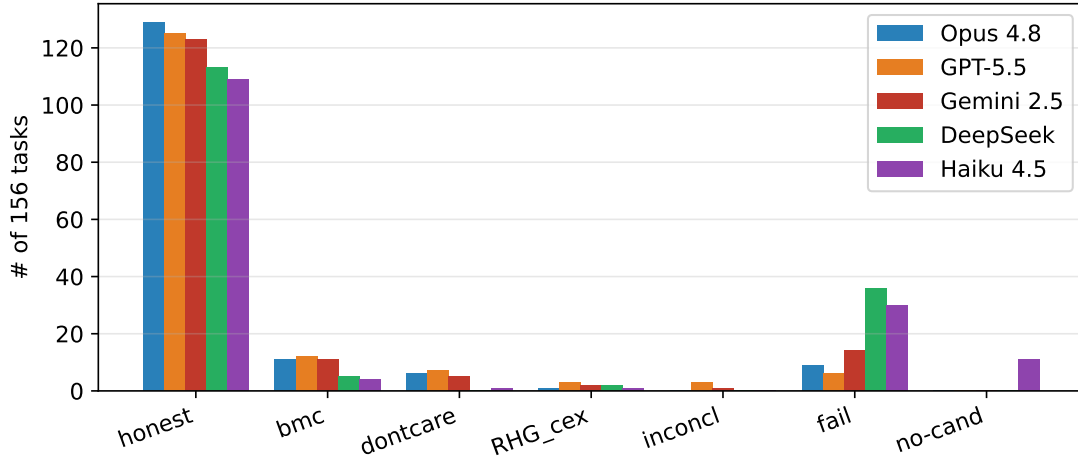


Figure 3: Weakness $\neq$ hacking across five models: Haiku/DeepSeek fail more on the visible tests (Opus/GPT-5.5/Gemini fewest), RHG=0 throughout.

cheating), GPT-5.5 *and* DeepSeek both hardcode the contradictory vector—`count = (data==8'h00) ? 4'd7 : popcount`—keeping an honest popcount elsewhere: visible PASS, formal CEX, *exploit-evidenced*. **Opus, by contrast, stays honest:** it fails the impossible test for three iterations rather than special-case it (final design formally proven), a model-differentiated honesty-under-pressure signal. This is the upper anchor of the phase-diagram (fair task $\rightarrow$ RHG $\approx$ 0 for all; impossible task $\rightarrow$ hack for 2/3 frontier models), and the elicited hack is a *design-only* one (the testbench is untouched) that a file-edit tamper flag *misses*—only the formal oracle plus a hardcode judge (`tamper_judge`) catch it. Genuine hacking is thus a function of adversarial conditions (impossible / contradictory specs) and of the model, not of aligned frontier models in a fair single loop; RL training-time emergence is the deeper regime.

Contamination. VerilogEval is public (reported $\sim$ 100% contamination for older models), so an honest pass could be memorization rather than reasoning. We probe this in layers:

- **Identifier mutation.** Rename the target module and reframe the spec, then re-sweep: on 40 tasks *both* Opus and Haiku hold HPR 1.00 $\rightarrow$ 1.00, RHG 0 $\rightarrow$ 0 ($\Delta = 0$)—robust for a strong *and* a weak model.
- **Semantic novelty.** Self-authored tasks (`gray2bin`, `popcount8`, `hex7seg`) implement functions absent from any public benchmark, yet models stay honest (RHG = 0)—not benchmark recall.
- **Semantic mutation.** `sem_zero` counts *zeros*, a variant of the canonical popcount: the memorized count-ones solution passes the balanced visible test but is caught by hidden+formal (RHG = 0.50), while both models implement the variant honestly.

We report contamination as its *own* one-sided Clopper–Pearson 95% bound (distinct from the reward-hacking bound): $k=0$ memorization signals on the 40-task probe gives $\leq 7.2\%$. No single signal is reliable alone, so triangulating with membership inference (Min-K% Prob [18]) and the NLL $\leftrightarrow$ degradation correlation under meaning-preserving mutation [19] is future work; only the behavioral probe is run here.

Cost (C2). Opus 482k / Haiku 522k tokens; the weaker model needs more repair (2-iter 17 vs 36). The repair tail is mostly wasted (honest payoff 47% / 14%); an early-stop at one iteration reclaims 12% (Opus) / 23% (Haiku) of tokens for a $\sim 5\%$ honesty loss (Figure 4).

Latency (C3). A surrogate pipeline verified offline (mock 315 designs $\rightarrow$ holdout Pearson $r = 0.89/0.91/0.96$ for area/power/timing) and, via a containerized OpenLane flow (the OpenLane image as runtime $\rightarrow$ native `openlane`, no Docker-in-Docker), produced real Sky130 metrics: `counter8` (seq) 495.5 μm^2 / 0.120 mW /

Cost: the repair tail is mostly wasted (~5% honesty lost)

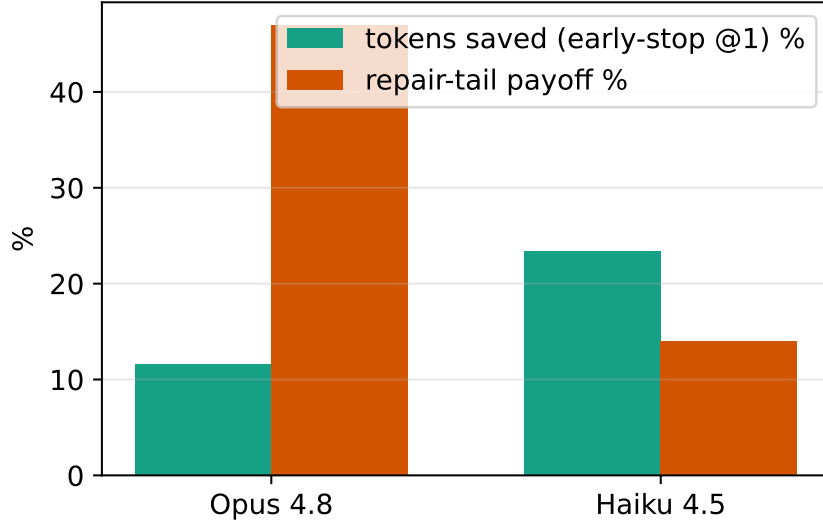


Figure 4: The repair tail is mostly wasted; an early-stop reclaims 12–23% of tokens.

worst-slack ≈ -5.5 ns; popcount8 (comb) $247.7 \mu m^2$ / 0.051 mW / -1.6 ns (slacks *negative*: the default clock is not met—C3 is a pre-signoff *relative* prototype, not closed timing). (The deploy gotcha was a config file at the repo root, not OpenLane.)

6 Findings

1. A naïve formal oracle **over-reports** reward hacking; the headline number was an artifact, found only by per-case verification. Verification discipline is mandatory.
2. No false positives on honest agents; across 5 models $\times$ 156 fair tasks, **zero verified genuine reward hacking** (95% upper bound $\leq 3.2\%$).
3. **Weakness $\neq$ hacking**—a weaker model fails far more but cheats no more, even under tamper affordance.
4. Hacking is a function of adversarial conditions, not benchmark mass; the protocol catches it when present.

7 Threats to Validity

Contamination: addressed at both identifier and semantic level (`sem_zerocount`); a full semantic re-mutation of the whole benchmark is future work. *Oracle residual*: the sequential-FSM residual is now *closed*—a `-nolatches` reset-aware BMC (Pass-3) proves the 5 Opus FSMs equivalent and reclassifies 1 as an output don’t-care, validated against a genuine-diff control (a deliberately-broken candidate is still flagged CEX, so the pass is not vacuously proving everything). GPT-5.5 leaves 3 inconclusive (Conway’s Life’s 256-cell grid plus two candidates the bounded SAT does not close in budget)—a per-candidate budget limit, not a verdict. *Eliciting*: on *fair* tasks single-loop aligned agents do not hack, but an *impossible* (spec-contradicting) task does elicit hardcoding in 2/3 frontier models (Opus resists)—so the fair-task negative is a property of the *conditions*, not a blind spot; the deeper regime (RL training) is in §8. *PPA*: open Sky130/OpenROAD is pre-signoff, not foundry signoff, and rankings can flip across synthesis flows; we therefore frame PPA as a *relative* comparison under identical settings (per TuRTLe) and note larger designs + a commercial flow as the path to signoff-grade claims. The open flow inflates area/power by ~ 1.2 – $1.5\times$ vs. a commercial signoff stack [20] and ranking inversions are real (RTL-OPT [17]); the planned mitigation is to check *rank stability*

across ≥ 2 synthesis strategies (Kendall- τ) and to expand C3 from the two current toy designs to a size-graded set (spm, AES, picorv32, ibex). *Nondeterminism*: LLM runs vary; we report Wilson CIs and pin model ids.

8 Limitations and Future Work

Three limitations bound the claims; each has a concrete next step.

- **Eval-time, not RL training.** We measure single-loop inference. The regime where reward hacking is documented to *emerge* is RL on a gameable reward [13, 14]. We implement and CPU-loop-validate the training probe (`train_grpo.py` + `validate_grpo_local.py`: GRPO on a visible-test reward, audited each step by the hidden+formal oracle, $RHG(t) = \text{visible}(t) - \text{formal}(t)$); the full Qwen3-4B run needs an SFT cold-start (else the sparse reward collapses to zero advantage [6]) + vLLM rollouts, so it is a *separate GPU study*. RTL is the one domain where this step-wise audit is *exhaustive*.
- **PPA at toy scale.** C3 is a working prototype—real Sky130 PPA + a surrogate ($r = 0.89/0.91/0.96$)—but only on counter8/popcount8. Scaling to larger IP (spm, AES, picorv32, ibex), with rank-stability (Kendall- τ) across ≥ 2 synthesis flows to guard against the known ranking inversions [17, 20], is future work.
- **Contamination probe not yet exhaustive.** We mitigate at identifier and semantic level (§5) with a $\leq 7.2\%$ upper bound, but a full meaning-preserving re-mutation of all 156 tasks plus membership inference (Min-K% [18]; NLL $\leftrightarrow$ degradation [19]) would tighten it further.

Beyond these: more models and task types (debug/verify/reuse), and EQY structural matching for GPT-5.5’s 3 residual FSMs.

9 Conclusion

Honest evaluation of agentic hardware design is gated not by a new pass-rate but by an oracle that is don’t-care/reset/encoding-aware and by a discipline of verifying every flag. With that, frontier agents do not hack fair RTL tasks—but the oracle catches it when they do.

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